STRATEGIC ANALYSIS & IMPLEMENTATION FRAMEWORK

Beverly Index
Veterinary Applications

NOST/NOSS Framework for Companion Animal Mental Health

Transforming companion animal mental health assessment through objective, technology-enabled screening—leveraging observable sleep behaviors as proxies for dream-state analysis in dogs, cats, and horses.

$62B+	94M	68%	0
GLOBAL VET MARKET	U.S. PET HOUSEHOLDS	SENIOR DOGS WITH CCD	FDA-APPROVED CAT TREATMENTS

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Executive Summary

This document presents a comprehensive analysis of veterinary applications for the Beverly Index technology platform, specifically examining how the Narrative Oneiric State Taxonomy (NOST) and Narrative Oneiric State Spectra (NOSS) frameworks can be adapted for companion animal mental health assessment.

The veterinary healthcare market represents a **$62+ billion global opportunity**, with companion animal care driving significant growth. Mental health conditions in pets—including cognitive dysfunction syndrome (CDS), anxiety disorders, and trauma-related conditions—affect millions of animals annually, yet current diagnostic tools rely heavily on subjective behavioral observation and owner-reported symptoms.

Beverly Index proposes a paradigm shift: leveraging observable sleep behaviors as proxies for dream-state analysis, creating the **first objective, technology-enabled assessment system** for companion animal cognitive and emotional health.

Key Market Findings

- ✓ 94 million U.S. households own pets; 70% of American homes have at least one pet
- ✓ 28-68% of senior dogs (11-16 years) show cognitive dysfunction signs
- ✓ 55-80% of senior cats (11-20 years) exhibit cognitive decline
- ✓ 5-17% of dogs affected by canine PTSD, higher in rescue populations
- ✓ No current FDA-approved treatments for feline cognitive dysfunction

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Scientific Foundation: Animal Dreaming

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Evidence for Animal Dreams

Research confirms that mammals, particularly dogs and cats, experience REM (Rapid Eye Movement) sleep patterns remarkably similar to humans. During REM sleep, animals exhibit behaviors strongly suggestive of dream activity:

- **MIT researchers (2001)** demonstrated rats replay maze-running patterns during REM sleep, with neural activation matching waking experiences
- **French neuroscientist Michel Jouvet's** landmark studies showed cats with disabled pons began 'acting out' hunting behaviors during sleep
- **Harvard researcher Dr. Deirdre Barrett** confirms: "REM sleep looks very similar across all mammals," supporting the hypothesis that dogs dream about daily experiences—including their owners
- **Dogs exhibit predictable REM cycles:** smaller dogs dream more frequently (every 10 minutes) with shorter duration, while larger dogs dream less often (every 60-90 minutes) with longer episodes

Observable Sleep Behaviors

Unlike humans, animals cannot verbally report dream content. However, observable sleep behaviors provide rich diagnostic data:

BEHAVIOR CATEGORY	OBSERVABLE SIGNS	POTENTIAL SIGNIFICANCE
Motor Activity	Leg twitching, running motions, tail wagging	Dream content involving movement, pursuit, play
Vocalizations	Whimpering, barking, growling, crying	Emotional valence of dream (positive/negative)
Eye Movement	Rapid movement behind closed lids	REM phase confirmation, visual dream processing

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Autonomic Response	Breathing rate changes, heart rate variability	Stress/arousal levels during sleep
Sleep Disruption	Frequent waking, restlessness, inability to settle	Sleep quality, potential anxiety or pain

The Beverly Index Adaptation Hypothesis

While animals cannot provide verbal dream narratives, the Beverly Index system can be adapted through a multi-modal approach:

- **Observable Sleep Behavior Analysis:** Pattern recognition of motor, vocal, and physiological sleep behaviors
- **Owner-Reported Observations:** Structured questionnaires capturing sleep behavior patterns over time
- **Waking Behavioral Correlation:** Linking daytime behaviors (anxiety, cognition, social interaction) with sleep patterns
- **Physiological Monitoring:** Integration with wearable devices tracking heart rate, movement, and sleep cycles

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Target Species Analysis

The veterinary application of NOST/NOSS focuses on domestic companion animals with confirmed REM sleep activity and significant mental health conditions affecting quality of life.

Canines (Dogs)

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PRIMARY TARGET

U.S. Households	**~65 million (48%)**
CCD at 11-12 years	**28%**
CCD at 15-16 years	**68%**
PTSD Prevalence	**5-17%**
Annual Vet Spending	**$350-500/dog**

Felines (Cats)

SECONDARY TARGET

U.S. Households	**~47 million (32%)**
FCD at 11-15 years	**28-55%**
FCD at 15+ years	**50-80%**
FDA-Approved Treatments	**None**
Key Opportunity	**Major diagnostic gap**

Equines (Horses)

TERTIARY TARGET

Sleep Pattern	**Limited REM**
Investment Level	**High per-animal**

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Key Conditions	**Anxiety, stereotypies**
Applications	**Rescue trauma, performance**
Timeline	**Phase 4+**

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NOST/NOSS Framework Adaptation

Adapted NOST Categories for Companion Animals

The human NOST framework's seven categories can be adapted for veterinary application, with modified assessment methodologies:

NOST CATEGORY	HUMAN ASSESSMENT	VETERINARY ADAPTATION
I. Lucidity/Control	Dream awareness, narrative control	Sleep quality metrics, startle responses, sleep cycle regularity
II. Emotional Valence	Emotional content analysis	Sleep vocalizations (positive vs. distress), waking affect correlation
III. Memory Integration	Temporal/episodic markers	Learned behavior retention, recognition tests, daily experience replay
IV. Sensory/Motor	Sensory detail, movement	Sleep motor activity, twitching patterns, breed-specific movements
V. Social/Relational	Relationship processing	Attachment behaviors, separation response, social hierarchy
VI. Anxiety/Threat	Threat perception analysis	Fear responses, startle threshold, environmental reactivity

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VII. Trauma/CTE	TBI/trauma markers	Trauma history correlation, abuse recovery patterns, rescue dog profiles

Adapted NOSS Classification System

The NOSS classification provides 49 distinct states through category combinations. For veterinary use, we propose a simplified 16-state model focusing on clinically actionable categories:

Low Anxiety	Moderate Anxiety	High Anxiety
High Cognition		
VET-A1: Optimal Healthy baseline		
VET-A2: Vigilant Monitor		
VET-A3: Stressed Intervention needed		
Moderate Cognition		
VET-B1: Aging Normal decline		
VET-B2: Mild CDS Early intervention		
VET-B3: Anxious Decline Dual focus		
Low Cognition		
VET-C1: Advanced CDS Supportive care		
VET-C2: Complex CDS Specialist referral		
VET-C3: Severe CDS Palliative focus		

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Veterinary-Specific NOSS Extensions

Additional veterinary-specific states address conditions unique to animal populations:

- **VET-T1:** Acute Trauma (Recent event, active symptoms)
- **VET-T2:** PTSD Pattern (Chronic trauma response, trigger sensitivity)
- **VET-T3:** Recovery Trajectory (Improving trauma indicators)
- **VET-P1:** Pain-Related Sleep Disruption
- **VET-P2:** Chronic Pain Adaptation
- **VET-S1:** Separation Anxiety Pattern
- **VET-S2:** Environmental Anxiety Pattern

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Clinical Applications

Canine Cognitive Dysfunction (CCD) Assessment

Canine Cognitive Dysfunction is the veterinary equivalent of Alzheimer's disease, sharing pathological hallmarks including beta-amyloid accumulation and neuronal death. Current diagnosis relies on owner observation using **DISHAAL criteria**:

- **D**isorientation in familiar environments
- **I**nteraction changes with family/other pets
- **S**leep/wake cycle disturbances
- **H**ouse-soiling (loss of training)
- **A**ctivity level changes

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- **A**nxiety increase
- **L**earning/memory deficits

Beverly Index CCD Assessment Enhancement:

- Quantitative sleep pattern analysis replacing subjective 'sleep disturbance' observations
- Longitudinal tracking enabling detection of subtle early-stage changes
- Correlation with CADES scores for validation
- Treatment response monitoring (enrichment, diet, medication)

Feline Cognitive Dysfunction (FCD) Assessment

Feline cognitive dysfunction presents unique challenges:

- No FDA-approved treatments (Anipryl approved only for dogs)
- Cats mask symptoms as survival mechanism
- VISHDAAL criteria less validated than canine equivalents
- Limited clinical research compared to canine populations

Beverly Index Opportunity: First systematic, technology-enabled FCD screening tool, potentially accelerating pharmaceutical development by providing objective endpoints.

Canine PTSD & Anxiety Disorders

5-17% of dogs experience PTSD, with higher prevalence in rescue populations, military/police working dogs, and disaster survivors. Current symptoms tracked:

- Hypervigilance and startle responses
- Avoidance of specific people, places, or situations
- Sleep disturbances and nightmares

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- Aggression or fearfulness
- Changes in eating/sleeping patterns

Beverly Index PTSD Applications:

- Pre-adoption screening for rescue organizations
- Treatment progress monitoring for behavioral therapy
- Trigger identification through sleep pattern analysis
- Return-to-duty assessment for working dogs

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Market Opportunity

Global Veterinary Healthcare Market

MARKET METRIC	VALUE
Global Veterinary Healthcare Market (2025)	**$62.28 billion**
Projected Market (2030)	$86.66 billion (6.83% CAGR)
U.S. Pet Industry Total (2025)	**$157+ billion**
U.S. Veterinary Care Segment (2025)	$41.4 billion
Global Pet Care Market (2032)	$427.75 billion projected
Diagnostics Segment Growth	**7.33% CAGR** (fastest growing)

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Revenue Model

Clinical Assessment License

$500-2,000/month

Veterinary clinics subscription

Per-Assessment Fee

$25-75/assessment

Pay-per-use clinics

Research/Pharma License

$50,000-200,000/year

Drug companies, universities

Insurance API Access

Enterprise pricing

Pet insurance companies

Shelter Volume License

Discounted subscription

Rescue organizations

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Implementation Strategy

PHASE 1 | MONTHS 1-18

Canine Cognitive Dysfunction Validation

Partner with 3-5 veterinary neurology/behavior specialists. Develop standardized sleep behavior observation protocol. Create owner-reported questionnaire for home observations. Correlate Beverly Index scoring with CADES/CCDR scales. Publish validation study in veterinary journal. **Target: 500+ assessments, correlation coefficient >0.7 with CADES.**

PHASE 2 | MONTHS 12-30

Canine Anxiety/PTSD Applications

Partner with rescue organizations (ASPCA, Humane Society). Develop trauma-specific assessment protocols. Pilot pre-adoption screening program. Track adoption success rates with Beverly Index screening. **Target: 1,000+ rescue dog assessments, improved adoption outcomes.**

PHASE 3 | MONTHS 24-42

Feline Extension

Adapt canine protocols for feline sleep patterns. Partner with feline-specialist practices. Develop FCD-specific NOSS classifications. Pursue pharmaceutical partnership for FCD drug development. **Target: First technology-enabled FCD screening tool.**

PHASE 4 | MONTHS 36-60

Market Expansion

Consumer wearable integration (Whistle, Fi, etc.). Pet insurance partnerships. International market entry (UK, EU, Asia). Equine pilot program.

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Technical Considerations

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Data Collection Methods

Without verbal narrative input, the veterinary Beverly Index relies on multi-modal data:

Owner-Reported Observations:

- Structured questionnaires administered via mobile app or web portal
- Video capture of sleep behaviors for AI analysis
- Daily logs of waking behaviors correlating with sleep patterns

Veterinary Clinical Assessment:

- Standardized sleep observation protocols
- Cognitive testing battery (adapted CADES items)
- Behavioral assessment in controlled environment

Wearable Device Integration:

- Activity/sleep trackers providing objective movement data
- Heart rate variability during sleep
- GPS for wandering/disorientation tracking

Validation Requirements

$r \geq 0.70$

CADES CORRELATION

$ICC \geq 0.80$

TEST-RETEST RELIABILITY

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$\kappa \geq 0.70$

INTER-RATER RELIABILITY

$\geq 80\%$

SENSITIVITY (CCD)

$\geq 75\%$

SPECIFICITY

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Competitive Landscape

Current Assessment Tools

TOOL/METHOD	DESCRIPTION	LIMITATIONS
CADES Scale	13-item owner questionnaire for CCD	Subjective, retrospective, no objective measures
CCDR Scale	Cognitive dysfunction rating scale	Owner-reported, variable interpretation
MRI Imaging	Brain atrophy visualization	Expensive, requires anesthesia, late-stage detection
Behavioral Observation	Clinical assessment by veterinarian	Inconsistent, time-limited snapshot

Beverly Index Competitive Advantages

- **First technology-enabled, objective** cognitive/emotional assessment
- **Continuous monitoring** vs. point-in-time assessment
- **Earlier detection** through subtle pattern recognition
- **Treatment response tracking** with quantitative metrics
- **Integration** with existing wearable ecosystem
- **Scalable platform** applicable across species

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Conclusion & Recommendations

Strategic Recommendation

The veterinary application of Beverly Index technology represents a significant market opportunity with strong scientific foundation. We recommend proceeding with development based on:

- Strong scientific evidence for animal dreaming paralleling human REM sleep
- Large, underserved market for pet mental health assessment
- Clear diagnostic gaps in cognitive dysfunction and trauma assessment
- Growing pet healthcare spending and pet 'humanization' trends
- Potential pharmaceutical partnerships for drug development endpoints

Priority Actions

1. **Establish veterinary advisory board** (neurology, behavior, geriatric specialists)
2. **Develop canine sleep behavior observation protocol** with standardized scoring
3. **Initiate CCD validation pilot** with 3-5 specialty practices
4. **Explore wearable device partnerships** for objective data integration
5. **File provisional patent** for veterinary NOST/NOSS applications

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The Beverly Index Veterinary Vision

Transforming companion animal mental health through objective, technology-enabled assessment—because every pet deserves the same quality of cognitive care as their human family.

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Confidential | January 2026 | Version 1.0

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